

- Heat-shock proteins are found in plants, animals, and microorganisms.
- heavy chain** One of the two types of polypeptide chains that make up an antibody molecule and B cell receptor; consists of a variable region, which contributes to the antigen-binding site, and a constant region.
- helicase** An enzyme that untwists the double helix of DNA at replication forks, separating the two strands and making them available as template strands.
- helper T cell** A type of T cell that, when activated, secretes cytokines that promote the response of B cells (humoral response) and cytotoxic T cells (cell-mediated response) to antigens.
- hemocoel** A body cavity lined by tissue derived from mesoderm and by tissue derived from endoderm.
- hemoglobin** (hē'-mō-glō'-bin) An iron-containing protein in red blood cells that reversibly binds oxygen.
- hemolymph** (hē'-mō-limf') In invertebrates with an open circulatory system, the body fluid that bathes tissues.
- hemophilia** (hē'-muh-fil'-ē-uh) A human genetic disease caused by a sex-linked recessive allele resulting in the absence of one or more blood-clotting proteins; characterized by excessive bleeding following injury.
- hepatic portal vein** A large vessel that conveys nutrient-laden blood from the small intestine to the liver, which regulates the blood's nutrient content.
- herbivore** (hur'-bi-vōr') An animal that mainly eats plants or algae.
- herbivory** A +/- ecological interaction in which an organism eats part of a plant or alga.
- heredity** The transmission of traits from one generation to the next.
- hermaphrodite** (hur-maf'-ruh-dīt') An individual that functions as both male and female in sexual reproduction by producing both sperm and eggs.
- hermaphroditism** (hur-maf'-rō-dī-tizm) A condition in which an individual has both female and male gonads and functions as both a male and a female in sexual reproduction by producing both sperm and eggs.
- heterochromatin** (het'-er-ō-krō'-muh-tin) Eukaryotic chromatin that remains highly compacted during interphase and is generally not transcribed.
- heterochrony** (het'-uh-rok'-ruh-nē) Evolutionary change in the timing or rate of an organism's development.
- heterocyst** (het'-er-ō-sist) A specialized cell that engages in nitrogen fixation in some filamentous cyanobacteria; also called a heterocyte.
- heterokaryon** (het'-er-ō-kār'-ē-un) A fungal mycelium that contains two or more haploid nuclei per cell.
- heteromorphic** (het'-er-ō-mōr'-fik) Referring to a condition in the life cycle of plants and certain algae in which the sporophyte and gametophyte generations differ in morphology.
- heterosporous** (het-er-os'-pōr-us) Referring to a plant species that has two kinds of spores: microspores, which develop into male gametophytes, and megaspores, which develop into female gametophytes.
- heterotroph** (het'-er-ō-trōf) An organism that obtains organic food molecules by eating other organisms or substances derived from them.
- heterozygote** An organism that has two different alleles for a gene (encoding a character).
- heterozygote advantage** Greater reproductive success of heterozygous individuals compared with homozygotes; tends to preserve variation in a gene pool.
- heterozygous** (het'-er-ō-zī'-gus) Having two different alleles for a given gene.
- hibernation** A long-term physiological state in which metabolism decreases, the heart and respiratory system slow down, and body temperature is maintained at a lower level than normal.
- high-density lipoprotein (HDL)** A particle in the blood made up of thousands of cholesterol molecules and other lipids bound to a protein. HDL scavenges excess cholesterol.
- hindbrain** One of three ancestral and embryonic regions of the vertebrate brain; develops into the medulla oblongata, pons, and cerebellum.
- histamine** (his'-tuh-mēn) A substance released by mast cells that causes blood vessels to dilate and become more permeable in inflammatory and allergic responses.
- histogram** A variant of a bar graph that is made for numeric data by first grouping, or "binning," the variable plotted on the x-axis into intervals of equal width. The "bins" may be integers or ranges of numbers. The height of each bar shows the percent or number of experimental subjects whose characteristics can be described by one of the intervals plotted on the x-axis.
- histone** (his'-tōn) A small protein with a high proportion of positively charged amino acids that binds to the negatively charged DNA and plays a key role in chromatin structure.
- histone acetylation** (his'-tōn) The attachment of acetyl groups to certain amino acids of histone proteins.
- HIV (human immunodeficiency virus)** The infectious agent that causes AIDS. HIV is a retrovirus.
- holdfast** A rootlike structure that anchors a seaweed.
- homeobox** (hō'-mē-ō-boks') A 180-nucleotide sequence within homeotic genes and some other developmental genes that is widely conserved in animals. Related sequences occur in plants and yeasts.
- homeostasis** (hō'-mē-ō-stā'-sis) The steady-state physiological condition of the body.
- homeotic gene** (hō-mē-o'-tik) Any of the master regulatory genes that control placement and spatial organization of body parts in animals, plants, and fungi by controlling the developmental fate of groups of cells.
- hominin** (hō'-mi-nin) A group consisting of humans and the extinct species that are more closely related to us than to chimpanzees.
- homologous chromosomes (or homologs)** (hō-mol'-uh-gus) A pair of chromosomes of the same length, centromere position, and staining pattern that possess genes for the same characters at corresponding loci. One homologous chromosome is inherited from the organism's father, the other from the mother. Also called a homologous pair.
- homologous pair** See homologous chromosomes.
- homologous structures** (hō-mol'-uh-gus) Structures in different species that are similar because of common ancestry.
- homologs** See homologous chromosomes.
- homology** (hō-mol'-ō-jē) Similarity in characteristics resulting from a shared ancestry.
- homoplasy** (hō'-muh-play'-zē) A similar (analogous) structure or molecular sequence that has evolved independently in two species.
- homosporous** (hō-mos'-puh-rus) Referring to a plant species that has a single kind of spore, which typically develops into a bisexual gametophyte.
- homozygote** An organism that has a pair of identical alleles for a gene (encoding a character).
- homozygous** (hō'-mō-zī'-gus) Having two identical alleles for a given gene.
- horizontal gene transfer** The transfer of genes from one genome to another through mechanisms such as transposable elements, plasmid exchange, viral activity, and perhaps fusions of different organisms.
- hormone** In multicellular organisms, one of many types of secreted chemicals that are formed in specialized cells, travel in body fluids, and act on specific target cells in other parts of the organism, changing the target cells' functioning.
- hornwort** A small, herbaceous, nonvascular plant that is a member of the phylum Anthocerotophyta.
- host** The larger participant in a symbiotic relationship, often providing a home and food source for the smaller symbiont.
- host range** The limited number of species whose cells can be infected by a particular virus.
- Human Genome Project** An international collaborative effort to map and sequence the DNA of the entire human genome.
- human immunodeficiency virus (HIV)** The infectious agent that causes AIDS (acquired immunodeficiency syndrome). HIV is a retrovirus.
- humoral immune response** (hyū'-mer-ul) The branch of adaptive immunity that involves the activation of B cells and that leads to the production of antibodies, which defend against bacteria and viruses in body fluids.
- humus** (hyū'-mus) Decomposing organic material that is a component of topsoil.
- Huntington's disease** A human genetic disease caused by a dominant allele; characterized by uncontrollable body movements and degeneration of the nervous system; usually fatal 10 to 20 years after the onset of symptoms.
- hybrid** Offspring that results from the mating of individuals from two different species or from two true-breeding varieties of the same species.
- hybrid zone** A geographic region in which members of different species meet and mate, producing at least some offspring of mixed ancestry.
- hybridization** In genetics, the mating, or crossing, of two true-breeding varieties.
- hydration shell** The sphere of water molecules around a dissolved ion.
- hydrocarbon** An organic molecule consisting only of carbon and hydrogen.